🚀 Capstone Idea Evaluation

Here is a concise breakdown of your five capstone ideas, evaluated for efficiency and impact.

+4

🟢 Idea 1: Road-Surface Costmap Generator

- **Beneficial?** ⭐ ⭐ ⭐ ⭐ ⭐ Prevents expensive hardware damage to autonomous fleets by mapping 3D terrain.

  +1

- **Worth Expenses?** 💰 Yes. Uses standard RGB-D cameras to save thousands in vehicle repairs.

  +1

- **Applicable?** 🛠️ Highly. Integrates directly into existing ROS 2 navigation stacks.

- **Industry Need?** ⏳ **Critical.** Essential for the transition from "perfect lab" driving to "real-world" urban roads.

  +1

🔵 Idea 2: Predictive "Risk-Field" Navigation

- **Beneficial?** ⭐ ⭐ ⭐ ⭐ Solves the "robot gridlock" problem in crowded areas.

  +1

- **Worth Expenses?** 💰 High. Requires precise LiDAR and heavy math processing, but enables Level 4 autonomy.

  +1

- **Applicable?** 🛠️ Moderate. Requires complex probabilistic tuning to feel "natural" to humans.

  +1

- **Industry Need?** 🚶 **High.** Necessary for sidewalks, airports, and hospital delivery bots.

🟡 **Idea 3: "Guide-Me" Smart Library Assistant**

- **Beneficial?** ⭐ ⭐ ⭐ Improves indoor navigation in massive, complex buildings.

- **Worth Expenses?** 💰 Moderate. The value lies in the software integration (V2I) rather than the robot itself.

  +1

- **Applicable?** 🛠️ High. Uses established IoT protocols like MQTT and web dashboards.

  +1

- **Industry Need?** 📚 **Niche.** Very helpful for large institutions but limited to specific indoor layouts.

  +1

🟠 Idea 4: "Micro-Scout" Yield Estimator

- **Beneficial?** ⭐⭐⭐⭐⭐ Automates labor-intensive crop counting and health tracking.

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- **Worth Expenses?** 💰 **Best Value.** Uses a cheap RC chassis and Raspberry Pi to provide high-value data.

  +1

- **Applicable?** 🛠️ High. Edge AI (YOLO) is well-supported for these specific use cases.

  +1

- **Industry Need?** 🌿 **Very High.** Solves a major "blind spot" in indoor vertical farming.

  +1

🔴 Idea 5: "Shepherd" Swarm for Crowd Control

- **Beneficial?** ⭐⭐⭐⭐ Enhances public safety during life-threatening crush events.

  +1

- **Worth Expenses?** 💰 Low. Requires *multiple* robots (a swarm) to be effective, increasing costs.

  +1

- **Applicable?** 🛠️ Difficult. Managing multiple autonomous

units in chaotic crowds is technically risky.

+1

- **Industry Need? 🏟️ High (Emergent).** High interest from event planners for "dynamic architecture."

  +1

Would you like me to create a comparison matrix for the hardware costs of these five ideas? 📊